VISUALIZATION

Visualization is a way of representing. Data visualization is representing of data in a graphical or pictorial format for better understanding of data. Visualization gives a good idea of data and the trends in it.

In python, data visualization has multiple libraries like matplotlib, seaborn, plotly etc.

Let us dive deeper into Bokeh and Pandas

BOKEH

Bokeh is a Python library for creating interactive visualizations for modern web browsers. It helps you build beautiful graphics, ranging from simple plots to complex dashboards with streaming datasets. With Bokeh, you can create JavaScript-powered visualizations without writing any JavaScript yourself.

Bokeh at a Glance

Flexible

Bokeh makes it simple to create common plots, but also can handle custom or specialized use-cases.

Interactive

Tools and widgets let you and your audience probe "what if" scenarios or drill-down into the details of your data.

Shareable

Plots, dashboards, and apps can be published in web pages or Jupyter notebooks.

Productive

Work in Python close to all the PyData tools you are already familiar with.

Powerful

You can always add custom JavaScript to support advanced or specialized cases.

Open Source

Everything, including the Bokeh server, is BSD licensed and available on GitHub.

Scatter Charts

Scatter plots are a plot of each data point in the data. You use them to plot two numeric variables against one another, and each data point on the x-axis has a corresponding, individually plotted value on the y-axis. As a result, the plot looks like a bunch of scattered points.

Code snippet:

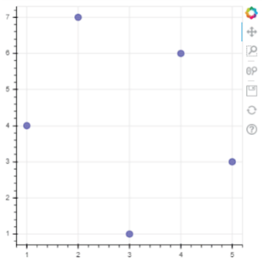
```
import pandas as pd
import pandas_bokeh
from bokeh.io import show, output_notebook
from bokeh.plotting import figure
import glob
pandas_bokeh.output_notebook()
pd.set_option('plotting.backend', 'pandas_bokeh')
```

BokehJS 2.1.1 successfully loaded.

Output:

Scatter Plot ¶

```
p = figure(plot_width = 400, plot_height = 400)
p.circle([1, 2, 3, 4, 5], [4, 7, 1, 6, 3], size = 10, color = 'navy', alpha = 0.5)
show(p)
```

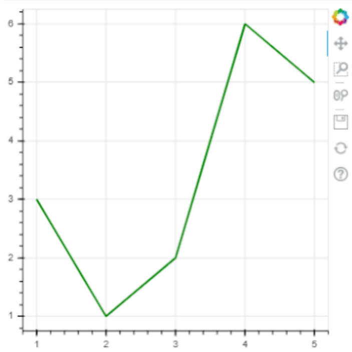


Line Chart

A line chart represents data as a series of points connected by a line. It is used to see trends in your data and track the way data changes over a period of time.A line plot can be drawn with the help of the line function in the plotting module of bokeh. Plotting contains all the graphs that can be plotted in Python bokeh.

Code snippet & output:

```
# create figure
p = figure(plot_width = 400, plot_height = 400)

# add a line renderer
p.line([1, 2, 3, 4, 5], [3, 1, 2, 6, 5],
        line_width = 2, color = "green")

# show the results
show(p)
```



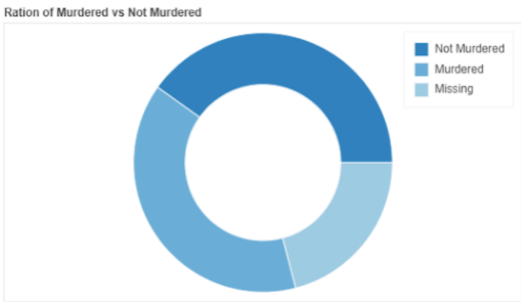
Pie Charts

A pie chart is a circular chart divided into slices to represent how much data belongs to a specific category. It is a quick and easy way to see the classes in your data and the percentage of the dataset they represent. Donut charts are like pie charts with a hole in the center.

Code snippet:

```python
from bokeh.transform import cumsum

fig = figure(plot_height=350, title="Ration of Murdered vs Not Murdered", toolbar_location=None,

tools="hover", tooltips="@Murdered: @Value", x_range=(-.5, .5))

fig.annular_wedge(x=0, y=1, inner_radius=0.15, outer_radius=0.25, direction="anticlock",

start_angle=cumsum('Angle', include_zero=True), end_angle=cumsum('Angle'),

line_color="white", fill_color='Color', legend='Murdered', source=df_mur)

fig.axis.axis_label=None

fig.axis.visible=False

fig.grid.grid_line_color = None

show(fig)
```

Output:



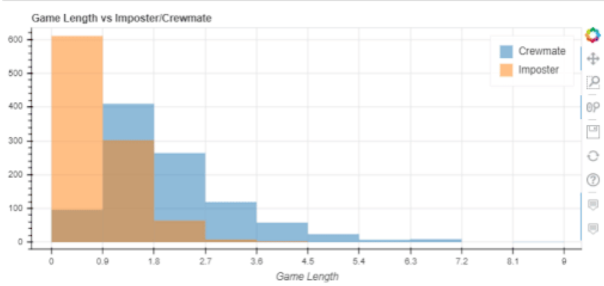
This shows above graph shows the values of Murdered and Not Murdered are close to each other, with a significant amount of values missing. Because of this, you cannot determine if the majority of the people were murdered or not.

Histogram

Histograms are used to plot numerical data according to the range they fall into. The data is plotted in bins or rectangles. The y-axis corresponds to the amount of data present in a specific range or at a certain point.

Code snippet & output:

```python
df_gm_te = pd.crosstab(df['Game Length'], df['Team'])
df_gm_te.plot_bokeh.hist(title='Game Length vs Imposter/Crewmate', figsize=(750, 350))
```



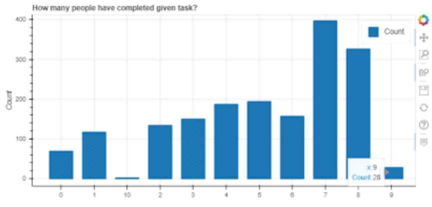
Figure(id = '2020', ...)

Bar Plot

While histograms plot numerical data distributions, bar plots represent the data distribution for categorical data. It also uses bins to plot the amount of data. When the bins are stacked on top of each other, it is called a stacked bar chart.

Code snippet & output:

```
df_tc = pd.DataFrame(df['Task Completed'].value_counts())[1:].sort_index().rename(columns={'Task Completed': 'Count'})
df_tc.plot_bokeh(kind='bar', y='Count', title='How many people have completed given task?', figsize=(750, 350))
```



Figure(id = '2250', ...)

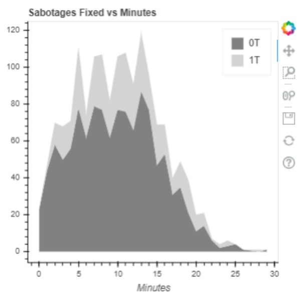
Area Plot

An area chart combines the line and bar chart when the area below a line is shaded. Using an area chart, you can see how the value of different groups changes over a period of time. The area chart has different baselines to show the vertical range of data.

Code snippet:

```
from bokeh.models import ColumnDataSource
from bokeh.plotting import figure, output_file, show

# data

df_min = pd.crosstab(df['Min'], df['Sabotages Fixed']).reset_index()
df_min = df_min.rename(columns={0.0:'0T', 1.0:'1T',
2.0:'2T',3.0:'3T',4.0:'4T',5.0:'5T'
})

# chart

names = ['0T','1T']
source = ColumnDataSource(data=dict(
x = df_min.Min,
y0 = df_min['0T'],
y1 = df_min['1T']
))

fig = figure(width=400, height=400, title='Sabotages Fixed vs Minutes')
fig.varea_stack(['y0','y1'], x='x', color=("grey", "lightgrey"),legend_label=names, source=source)
fig.grid.grid_line_color = None
fig.xaxis.axis_label='Minutes'
show(fig)
```

Output:



Pandas

Pandas is an open-source Python library used for data manipulation, analysis and cleaning. It provides fast and flexible tools to work with tabular data, similar to spreadsheets or SQL tables.

Pandas is widely used in data science and analytics due to its integration with libraries such as:

- NumPy: numerical operations
- Matplotlib and Seaborn: data visualization
- SciPy: statistical analysis
- Scikit-learn: machine learning workflows

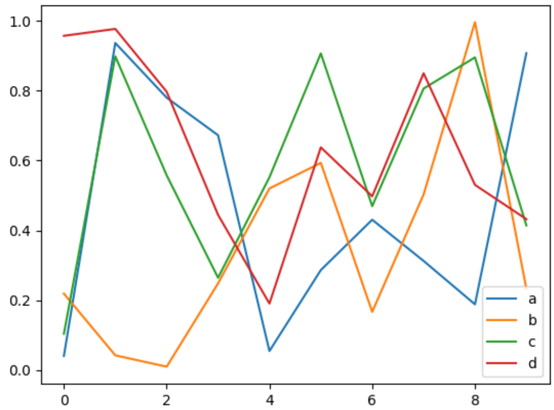
Line Plots

A Line plot is a graph that shows the frequency of data along a number line. It is best to use a line plot when the data is time series. It can be created using **Dataframe.plot()** function.

Code:

```
import numpy as np
import pandas as pd
df1 = pd.read_csv('df1', index_col=0)
df2 = pd.read_csv('df2')
df2.plot()
```

output:



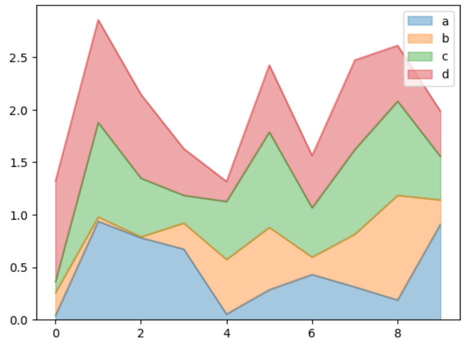
Area Plots:

Area plot shows data with a line and fills the space below the line with color. It helps see how things change over time. we can plot it using DataFrame.plot.area() function.

Code:

```
import numpy as np
import pandas as pd
df1 = pd.read_csv('df1', index_col=0)
df2 = pd.read_csv('df2')
df2.plot.area(alpha=0.4)
```

output:



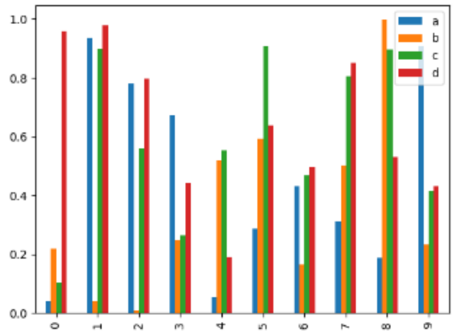
Bar Plots:

A bar chart presents categorical data with rectangular bars with heights or lengths proportional to the values that they represent. The bars can be plotted vertically or horizontally with DataFrame.plot.bar() function.

Code:

```
import numpy as np
import pandas as pd
df1 = pd.read_csv('df1', index_col=0)
df2 = pd.read_csv('df2')
df2.plot.bar()
```

output:



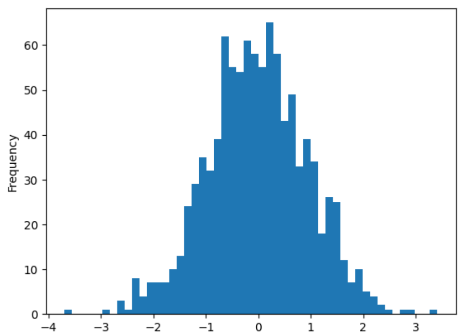
Histogram Plot:

Histograms help visualize the distribution of data by grouping values into bins. Pandas use **DataFrame.plot.hist()** function to plot histogram.

Code:

```
import numpy as np

import pandas as pd

df1 = pd.read_csv('df1', index_col=0)

df2 = pd.read_csv('df2')

df1['A'].plot.hist(bins=50)
```

output:



Scatter Plot:

Scatter plots are used when you want to show the relationship between two variables. They are also called correlation and can be created using DataFrame.plot.scatter() function.

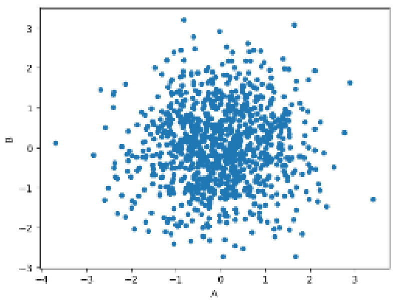
Code:

```
import numpy as np

import pandas as pd

df1 = pd.read_csv('df1', index_col=0)
```

df2 = pd.read_csv('df2')

df1.plot.scatter(x ='A', y ='B')

output:



Comparasion of Bokeh and Pandas

Bokeh and Pandas are both popular Python libraries but serve fundamentally different purposes in the data workflow. Understanding their differences can help you choose the right tool for your tasks. Below is a detailed comparison.

1. Purpose

- **Pandas:**
  - Mainly used for **data manipulation and analysis**.
  - Provides powerful data structures like DataFrame and Series for handling structured data.
  - Focused on **data cleaning, aggregation, transformation**, and exploration.
- **Bokeh:**
  - A **data visualization library** for creating interactive and web-ready plots.
  - Specializes in **interactive charts**, dashboards, and visual storytelling.
  - Focuses on presenting data rather than manipulating it.

2. Typical Use Cases

- Pandas:
  - Cleaning messy datasets.
  - Aggregating or summarizing data.
  - Preparing datasets for machine learning.
  - Quick analysis with simple visualization.
- Bokeh:
  - Creating interactive dashboards.
  - Producing plots for web applications.
  - Visual storytelling with dynamic updates.
  - Linking multiple plots with callbacks or widgets.

3. Strengths and Weaknesses

- **Pandas Strengths:**
  - Highly efficient for structured data manipulation.
  - Large ecosystem for scientific computing.

    - Simple for one-off analyses.
- **Pandas Weaknesses:**
    - Limited interactive visualization features.
    - Purely static plots (using Matplotlib for enhancement).
- **Bokeh Strengths:**
    - Native interactive plotting.
    - Supports real-time streaming data visualizations.
    - Easy conversion to standalone HTML or embedded web apps.
- **Bokeh Weaknesses:**
    - Not meant for heavy data computation.
    - Requires data preparation (often via Pandas or NumPy).

4. Core Functionality

Feature	Pandas	Bokeh
Data Storage	Yes (DataFrame, Series)	No, usually consumes data from Pandas
Data Analysis	Extensive (groupby, pivot, apply)	Limited (mainly visualization-based)
Visualization	Basic (plotting via df.plot())	Advanced (interactive, web-ready)
Interactivity	None	High (hover, zoom, sliders)
Output Format	Data analysis tables	HTML, JavaScript, interactive apps
Integration	Works with NumPy, Matplotlib	Can read Pandas DataFrames, integrates with Bokeh server or Flask/Django

5. Integration Workflow

A typical data analysis workflow might combine both tools:

```
import pandas as pd

from bokeh.plotting import figure, show

from bokeh.io import output_notebook

# Prepare and analyze data with Pandas

df = pd.read_csv("data.csv")

summary = df.groupby('category')['value'].mean()

# Create interactive plot with Bokeh using Pandas data

output_notebook()

p = figure(title="Average Values by Category", x_range=summary.index.tolist())

p.vbar(x=summary.index.tolist(), top=summary.values, width=0.5)

show(p)
```

Summary

- Use **Pandas** for **data analysis and manipulation**, especially large, structured datasets.
- Use **Bokeh** for **interactive and web-based visualizations**, often consuming pre-processed data from Pandas.
- Together, they complement each other in data science and analytics pipelines.